

POPRAD-TATRY, Slovakia

WMO: 119340

Lat: 49.074N

Lon: 20.241E

Elev: 718

StdP: 92.99

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-15.0	-20.6	0.7	-17.2	-17.9	0.8	-13.9	13.8	1.1	12.6	1.2	1.5	270	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	28.1	18.8	26.2	17.9	24.7	17.2	19.7	26.4	18.7	24.9	17.8	23.4	3.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.4	22.6	16.4	12.7	21.5	15.6	12.1	20.6	59.4	26.6	56.3	25.0	53.3	23.5	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.7	8.5	-21.5	31.1	3.4	1.6	-24.0	32.2	-26.0	33.2	-28.0	34.1	-30.5	35.3		
(5)			-21.8	21.4	3.4	1.0	-24.3	22.1	-26.2	22.7	-28.1	23.2	-30.6	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-4.1	-2.3	1.7	7.3	11.8	15.2	16.9	16.6	12.0	7.4	2.4	-2.9
	DBStd	8.54	5.33	5.30	4.44	3.99	3.47	3.27	3.04	3.04	3.43	4.21	4.73	5.08
	HDD10.0	1913	438	345	258	98	22	2	0	1	18	104	229	398
	HDD18.3	4228	697	578	516	332	203	103	64	71	191	339	477	657
	CDD10.0	775	0	0	1	15	78	157	215	206	79	22	2	0
	CDD18.3	48	0	0	0	0	1	9	20	17	1	0	0	0
	CDH23.3	664	0	0	0	2	24	133	247	236	21	1	0	0
	CDH26.7	131	0	0	0	0	2	22	53	53	2	0	0	0
Wind	WSAvg	3.3	3.4	3.4	3.9	3.5	3.3	3.2	3.2	2.8	3.0	3.0	3.2	3.2
Precipitation	PrecAvg	601	24	25	29	44	73	88	85	73	51	42	42	27
	PrecMax	997	77	85	87	118	157	213	220	186	119	188	135	85
	PrecMin	413	0	2	3	4	4	22	8	11	3	1	0	0
	PrecStd	116	16	17	17	25	33	36	46	38	29	37	28	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	11.1	16.3	22.2	25.9	28.9	30.7	30.9	25.8	21.8	15.1	9.2
		MCWB	5.1	6.2	8.0	12.8	15.9	19.2	20.1	19.1	17.0	13.9	10.7	6.7
	2%	DB	6.1	8.2	13.2	19.8	23.4	26.9	28.0	28.0	23.1	19.0	12.8	7.1
		MCWB	4.4	5.0	7.3	11.5	15.3	18.2	18.9	18.9	15.9	13.2	9.5	5.4
	5%	DB	4.5	6.6	10.9	17.2	21.6	24.8	26.1	25.9	21.2	16.9	11.0	5.2
		MCWB	2.8	3.8	6.2	10.2	14.3	17.3	18.0	17.9	14.7	12.2	8.4	3.7
	10%	DB	3.0	4.9	8.6	15.0	19.4	22.8	24.1	24.0	19.1	14.5	9.2	3.7
		MCWB	1.7	2.7	5.0	9.2	13.3	16.3	17.2	17.0	13.8	11.0	7.3	2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	7.0	9.2	13.6	17.7	20.7	21.0	20.7	18.1	15.0	11.1	7.2
		MCDB	7.2	9.7	14.1	20.4	23.8	26.6	27.2	27.5	24.3	19.5	14.0	8.9
	2%	WB	4.6	5.4	7.8	12.1	16.0	19.0	19.6	19.5	16.7	13.6	9.7	5.5
		MCDB	5.9	8.0	12.2	18.1	21.8	25.3	26.4	26.4	21.9	17.9	12.3	6.9
	5%	WB	3.0	4.0	6.6	10.9	14.8	17.9	18.7	18.5	15.5	12.5	8.7	3.8
		MCDB	4.2	6.0	10.2	16.2	20.3	23.5	24.7	24.8	20.1	16.3	10.8	4.9
	10%	WB	1.6	2.8	5.2	9.7	13.7	16.8	17.8	17.5	14.3	11.3	7.5	2.5
		MCDB	2.7	4.5	8.2	14.2	18.6	21.9	23.2	23.0	18.3	14.4	9.2	3.5
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.6	9.5	11.2	11.5	11.6	11.7	12.4	11.1	10.2	7.5	7.1
		MCDBR	6.8	10.0	13.5	15.4	15.2	15.2	15.4	14.6	13.8	9.6	7.4	
		MCWBR	5.5	7.1	8.6	8.5	8.2	8.1	7.5	7.5	8.2	8.5	6.8	6.0
	5% WB	MCDBR	6.4	8.6	12.0	13.7	13.7	13.7	13.7	14.3	12.9	12.7	8.6	6.7
		MCWBR	5.4	6.5	7.9	7.9	7.8	7.7	7.2	7.3	7.5	8.0	6.3	5.6
Clear-Sky Solar Irradiance	taub	0.288	0.315	0.362	0.424	0.418	0.420	0.434	0.430	0.396	0.360	0.328	0.288	
	taud	2.280	2.217	2.179	2.126	2.211	2.243	2.206	2.223	2.284	2.356	2.370	2.339	
	Ebn at Noon	760	821	837	820	843	842	823	806	797	764	708	712	
	Edh at Noon	72	98	121	142	136	133	136	128	109	86	67	60	
All-Sky Solar Radiation	RadAvg	0.92	1.68	2.86	4.14	4.99	5.34	5.17	4.76	3.36	2.05	1.05	0.70	
	RadStd	0.13	0.25	0.40	0.46	0.61	0.46	0.56	0.41	0.47	0.29	0.14	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	+0.45	N/A	N/A	N/A	+0.62	+0.54	N/A	N/A	+52	+15
(47)	Regional (45 neighbors)	+0.57	N/A	N/A	+0.95	+0.51	+0.45	-126	-181	+89	+40

Nomenclature: See separate page